

RESIDUAL SPECTRAL GRAPH LEARNING FOR CONNECTIVITY-LOSS ESTIMATION UNDER MULTI-EDGE DISRUPTIONS

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ABSTRACT. We estimate algebraic-connectivity loss after simultaneous road-network edge disruptions. The proposed model learns a bounded graph-neural correction to an interpretable first-order Fiedler approximation. We compare eight learned configurations and the analytical baseline under independent and spatially clustered failures, using graph-disjoint synthetic splits and zero-shot transfer to five OpenStreetMap (OSM) networks. Results are replicated with five seeds and uncertainty is computed by resampling seeds, not individual scenarios. On OSM, the residual model obtains mean absolute error (MAE) 0.097 (95% seed-bootstrap CI 0.091–0.102) for independent failures and 0.090 (0.071–0.124) for spatial failures. Relative to a direct GCN, paired MAE improvements are 0.0085 (0.0039–0.0131) and 0.0121 (0.0026–0.0247), respectively. Fiedler-node and coordinate ablations are inconclusive, indicating that the explicit residual prior provides the robust gain. Exact OSM eigendecomposition averages 5.09 ms per scenario; the first-order update requires 0.018 ms and learned inference 0.44–0.67 ms on a CPU. This is structural disruption analysis, not hazard or traffic-flow prediction. Code, cached networks, and complete result tables are available at github.com/lee-vtruong/ResiliRoad.

Keywords. Algebraic connectivity; Fiedler vector; graph neural network; residual learning; road-network disruption; OpenStreetMap.

1. INTRODUCTION

Removing a few links can change global road-network connectivity. Exact spectral recomputation is reliable but may be expensive in large scenario studies, while a local sensitivity is fast but ignores finite interactions. Graph neural networks (GNNs) can represent such interactions, yet direct end-to-end learning may discard useful mathematics. We therefore ask whether a GNN should predict connectivity loss directly or correct a spectral prior.

Our contributions are: (i) a residual spectral-GCN for multi-edge disruption; (ii) separation of an explicit spectral prior from Fiedler node features; (iii) independent and spatial-cluster failures on five OSM networks; (iv) direct GCN, Deep Sets, summary-MLP, feature, and residual ablations; and (v) seed-level paired uncertainty, runtime, and regime-specific error analysis.

We organize the experiment around four questions. **RQ1:** Does residual learning improve a direct GCN under graph-disjoint testing? **RQ2:** Does the advantage persist under zero-shot transfer and spatially correlated failures? **RQ3:** Are gains attributable to the Fiedler node coordinate, the spatial coordinates, or the explicit spectral prior? **RQ4:** What is the accuracy–runtime trade-off and in which graph regimes does the method fail?

2. RELATED WORK AND NOVELTY POSITIONING

Fiedler’s algebraic connectivity λ_2 characterizes connectedness and underlies spectral robustness analysis [1, 2]. Its eigenvector gives an edge-weight derivative, but summing derivatives is a local approximation and need not capture simultaneous finite deletions.

Road robustness has been studied under random, targeted, and localized link failures. Synthetic road graphs permit controlled topology experiments [3]; hazard-independent multi-link analysis also demonstrates that conclusions depend on disruption multiplicity [4]. Our target is structural connectivity, complementary to travel time, flow, or accessibility and not a substitute for them.

GCNs propagate local information over graph adjacency [5], whereas Deep Sets supplies a permutation-invariant baseline without message passing [6]. Physics-informed graph learning combines analytical structure and learned components [7]. We do not claim the first use of GNNs or spectral robustness. The novelty is the controlled, leakage-resistant comparison of node-level spectral information versus an explicit residual prior, across two failure processes and five unseen OSM networks, with paired seed-level inference and computational profiling.

3. FORMULATION AND MODELS

Let $G = (V, E, w)$ be connected and undirected, $L = D - A$ its combinatorial Laplacian, and $0 = \lambda_1 \leq \lambda_2 \leq \dots$ its eigenvalues.

Definition 3.1 (Relative connectivity loss). *For disrupted edges $S \subseteq E$,*

$$(1) \quad y(G, S) = \text{clip} \left(1 - \lambda_2(G - S) / \lambda_2(G), 0, 1 \right).$$

Disconnection therefore gives $y = 1$.

If u_2 is a unit Fiedler vector and λ_2 is simple, then $\partial \lambda_2 / \partial w_{ij} = (u_{2,i} - u_{2,j})^2$. Our analytical baseline is

$$(2) \quad \hat{y}_{\text{spec}} = \text{clip} \left(\frac{\sum_{(i,j) \in S} w_{ij} (u_{2,i} - u_{2,j})^2}{\lambda_2(G)}, 0, 1 \right).$$

The ScenarioGCN receives the damaged adjacency and node features comprising normalized original degree, disrupted-edge incidence, absolute normalized Fiedler coordinate, and two normalized spatial coordinates. Three 48-unit GCN layers are followed by mean-max pooling and a two-layer head. The residual variant predicts

$$(3) \quad \hat{y}_{\text{res}} = \text{clip}(\hat{y}_{\text{spec}} + \tanh r_\theta(G, S), 0, 1).$$

We compare direct and residual versions with all features, without Fiedler, and without coordinates. Additional baselines are a Deep Sets node encoder and an MLP using graph summary statistics. All learned models use the same split, AdamW, mean-squared error, at most 80 epochs, and validation early stopping.

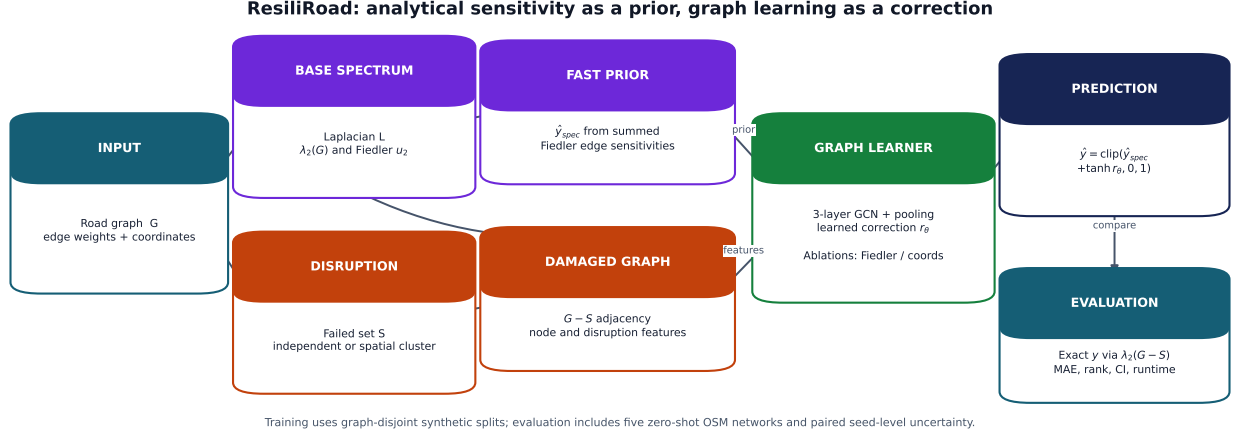


FIGURE 1. Method overview. The fast Fiedler sensitivity is an explicit prior; the GCN only learns its finite-disruption correction. Exact eigendecomposition is used to construct labels and evaluate accuracy, not as a model input.

3.1. Why residual learning? Equation (2) encodes the correct infinitesimal direction but is biased for large perturbations, eigenvalue crossings, and disconnections. A direct GCN must learn both this first-order structure and all higher-order effects. Residual learning instead allocates model capacity to the discrepancy. The tanh correction prevents unbounded updates, while clipping enforces the range implied by Equation (1). This construction does not guarantee an improvement: a severely shifted prior could mislead the learner, which is why the analytical, direct, and residual variants are all retained.

4. EXPERIMENTAL DESIGN

4.1. Synthetic data and failure processes. Connected random geometric graphs contain 35–65 nodes with inverse-length edge conductance. Each seed has 1,600 balanced scenarios (800 per failure mode): 1,120 train, 240 validation, and 240 test examples, split by base-graph identifier. No damaged version of a training graph can enter validation or test. Independent failure samples 1–8 edges uniformly. Spatial failure chooses a random epicentre and removes the requested number of edges whose midpoints are nearest to it.

The two mechanisms deliberately share the same failure-count distribution. Consequently, their difference isolates spatial dependence rather than greater damage magnitude. Edge weights and coordinates are computed on each intact base graph before deletion. The spectral prior is also computed from the intact graph, so its cost can be amortized when many scenarios share a network.

4.2. Five-network OSM transfer. OSMnx [8] downloaded 650-m drivable networks around five Vietnamese sites. Directed multigraphs were simplified, converted to undirected simple graphs, and restricted to their largest connected component. Conductance is $100/\text{length}_m$. Table 1 spans different sizes and densities. For every seed, site, and failure process, 100 scenarios were generated, giving 1,000 OSM scenarios per seed. OSM data are never used for training, early stopping, or model choice.

The five sites were selected before model comparison to cover a practical range of local street patterns rather than to maximize favourable performance. They span 48–206 nodes, 55–301 edges, and more than a threefold density range. The cached GraphML and download manifest preserve the exact graph instances because live OSM/Overpass queries may change over time.

TABLE 1. Cached OSM networks. Density is $2|E|/(|V|(|V| - 1))$.

Site	Nodes	Edges	Density
HCMUS (Ho Chi Minh City)	163	221	0.0167
VIASM (Hanoi)	206	301	0.0143
Da Nang	142	203	0.0203
Can Tho	114	178	0.0276
Da Lat	48	55	0.0488

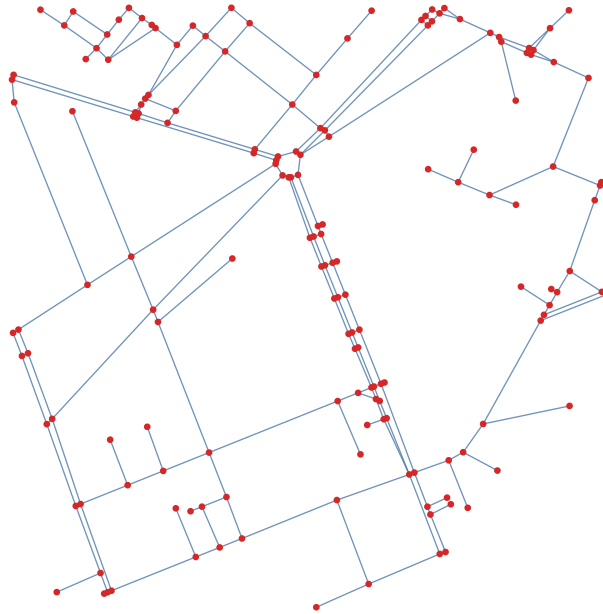


FIGURE 2. Example real-road graph used for zero-shot evaluation: the processed HCMUS OSM network with 163 nodes and 221 undirected edges. Red points denote intersections or terminal nodes and blue lines denote road links. The image shows the intact base graph; disruption scenarios remove selected links. Map data copyright OpenStreetMap contributors.

4.3. Statistics and runtime. Metrics are calculated within each seed. Reported 95% intervals resample the five seed-level values 10,000 times. Comparisons use within-seed paired differences; scenarios sharing a graph or seed are never treated as independent replicates. Runtime uses one PyTorch CPU thread on the same 16-logical-CPU Windows machine, and includes per-scenario prediction but excludes training and data loading.

For each seed and model we report MAE, RMSE, R^2 , and Spearman rank correlation. MAE is the primary calibration endpoint and Spearman correlation measures scenario-ranking quality. A confidence interval for a paired difference is obtained by first subtracting seed-matched metrics and then bootstrapping those five differences. This respects both shared scenarios and shared initialization/data-generation seeds. With only five independent replicates, intervals remain descriptive and no large-sample p -value is used.

4.4. Implementation. The implementation uses dense Laplacians because the evaluated graphs are small. Exact labels call a symmetric eigensolver on $G - S$. Neural inference is timed one scenario at a time to match the decision setting; batching could improve throughput but would confound comparison with the per-scenario eigensolver. PyTorch is restricted to one compute thread. The complete paper run trains eight configurations for each of five seeds, evaluates two synthetic and ten site-by-failure OSM test sets, and records timing beside every run.

5. RESULTS

TABLE 2. Synthetic test performance: seed mean MAE [95% seed-bootstrap CI] and Spearman correlation.

Method	Independent		Spatial cluster	
	MAE	ρ	MAE	ρ
Spectral first order	0.069 [0.048,0.095]	0.941	0.144 [0.102,0.183]	0.860
Summary MLP	0.065 [0.051,0.079]	0.593	0.108 [0.082,0.135]	0.610
Deep Sets	0.068 [0.058,0.078]	0.298	0.105 [0.085,0.126]	0.593
Direct GCN	0.077 [0.055,0.098]	0.587	0.081 [0.067,0.097]	0.549
Residual GCN	0.052 [0.036,0.070]	0.804	0.065 [0.049,0.079]	0.700

5.1. Synthetic graph-disjoint evaluation. The first-order model ranks independent synthetic failures extremely well but is less calibrated in MAE. Spatial clustering widens the gap between local sensitivity and finite damage. The residual model has the lowest core-model MAE in both cases, while retaining much of the prior’s ranking signal. Deep Sets underperforms GCN variants in ranking, suggesting that adjacency-aware message passing adds information beyond a permutation-invariant node aggregation.

TABLE 3. OSM zero-shot results: seed mean MAE [95% seed-bootstrap CI] and Spearman correlation.

Method	Independent		Spatial cluster	
	MAE	ρ	MAE	ρ
Spectral first order	0.502 [0.482,0.521]	0.391	0.650 [0.643,0.657]	0.324
Summary MLP	0.281 [0.240,0.320]	0.819	0.347 [0.269,0.425]	0.787
Deep Sets	0.114 [0.105,0.123]	0.765	0.143 [0.122,0.164]	0.789
Direct GCN	0.106 [0.100,0.112]	0.848	0.102 [0.077,0.148]	0.841
Residual GCN	0.097 [0.091,0.102]	0.903	0.090 [0.071,0.124]	0.896
Residual, no Fiedler	0.096 [0.090,0.101]	0.901	0.078 [0.067,0.090]	0.889
Residual, no coordinates	0.098 [0.092,0.102]	0.901	0.095 [0.073,0.122]	0.895

5.2. Zero-shot OSM transfer. Residual-full improves upon direct-full by paired MAE 0.0085 [0.0039,0.0131] for independent and 0.0121 [0.0026,0.0247] for spatial failures. On synthetic graphs the improvements are 0.0253 [0.0106,0.0400] and 0.0160 [0.0048,0.0241]. Thus the residual formulation is the stable contribution. In contrast, every paired MAE CI comparing residual-full with its no-Fiedler or no-coordinate variant contains zero. The spatial OSM point estimate even favours no-Fiedler. These features should be regarded as optional under the present power, not established necessities.

TABLE 4. Seed-paired MAE differences relative to residual-full. Positive means residual-full is better; intervals are 95% seed-bootstrap CIs.

Domain	Failure process	Direct GCN	Residual no Fiedler / no coordinates
Synthetic	Independent	0.0253 [0.0106,0.0400]	0.0001 [-0.0073,0.0074] / -0.0018 [-0.0073,0.0052]
Synthetic	Spatial	0.0160 [0.0048,0.0241]	-0.0027 [-0.0148,0.0072] / -0.0073 [-0.0192,0.0030]
OSM	Independent	0.0085 [0.0039,0.0131]	-0.0014 [-0.0034,0.0003] / 0.0001 [-0.0031,0.0033]
OSM	Spatial	0.0121 [0.0026,0.0247]	-0.0120 [-0.0507,0.0169] / 0.0047 [-0.0302,0.0435]

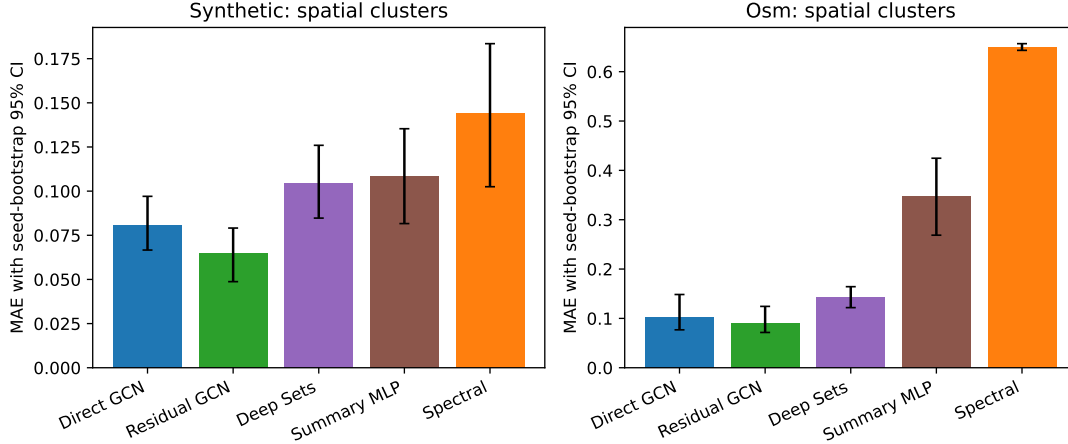


FIGURE 3. Spatial-cluster performance in the synthetic and OSM domains. Error bars are 95% intervals from seed-level bootstrap.

TABLE 5. Mean per-scenario runtime on OSM (milliseconds).

Method	ms	Method	ms
Exact eigendecomposition	5.094	Spectral first order	0.018
Summary MLP	0.442	Deep Sets	0.430
Direct GCN	0.502	Residual GCN	0.674

5.3. Runtime. The analytical approximation is about 287 times faster than exact recomputation; residual inference is about 7.6 times faster. These ratios are implementation- and graph-size-specific. Mean training time ranges from 60 s (summary MLP) to 150 s (slowest GCN ablation), and residual-full averages 121 s per seed on CPU.

5.4. Error regimes. For residual-full on pooled OSM failure modes, seed-mean MAE is 0.070 for connected damaged graphs and 0.112 after disconnection. By failed-edge count it is 0.090 (one), 0.116 (two–three), 0.089 (four–five), and 0.081 (six or more), so error is not monotone in disruption count. Site MAE is 0.041 Can Tho, 0.061 VIASM, 0.088 Da Nang, 0.089 HCMUS, and 0.189 Da Lat. The dense but much smaller Da Lat graph is hardest, warning against attributing error to density alone. No evaluated scenario clipped Equation (2) at one; consequently the requested clipped-versus-unclipped comparison is undefined rather than silently omitted. The poor spectral OSM MAE instead reflects large non-saturating overestimates.

TABLE 6. Residual-full OSM error slices, pooled over failure processes and summarized by seed. These are diagnostic strata, not independent experiments.

Slice	Group	MAE	Slice	Group	MAE
Connectivity	Connected	0.070	Area	Can Tho	0.041
	Disconnected	0.112		VIASM	0.061
Failed edges	1	0.090	Spectral clip	Da Nang	0.088
	2–3	0.116		HCMUS	0.089
	4–5	0.089		Da Lat	0.189
	6+	0.081		Clipped	0 cases

Disconnection is the clearest difficult regime. The peak at two–three failed edges occurs because failure count alone does not encode whether removed links form a cut. Likewise, density is confounded with graph size and street layout: the densest sample, Da Lat, is also the smallest and the hardest. Future experiments should vary size and density factorially rather than infer a causal density effect from five observational networks.

6. DISCUSSION AND LIMITATIONS

The explicit prior substantially improves a direct GCN even when the prior alone transfers poorly. It supplies useful rank structure, while the learned correction calibrates nonlinear finite-deletion effects. Deep Sets is competitive only in selected regimes and the summary MLP loses substantial accuracy, supporting the use of graph structure.

Five seeds permit uncertainty reporting but not high-powered claims about small feature effects. The OSM samples cover only 650-m neighbourhoods in Vietnam; Da Lat also differs in size as well as density. Spatial clusters are geometric proxies, not measured floods, landslides, or construction events. Algebraic connectivity ignores traffic demand, direction, capacity, travel time, and recovery. Future work should use larger networks, blocked geographic transfer, licensed hazard layers, and sparse/iterative eigensolvers.

7. CONCLUSION

Across both disruption processes and both domains, learning a correction to a Fiedler approximation is more reliable than direct GCN regression. Full feature ablation does not support strong claims for Fiedler node coordinates or spatial coordinates individually. Multi-area transfer, seed-paired statistics, runtime profiling, and explicit failure-regime analysis make the result suitable as a reproducible poster study while delimiting its real-world interpretation.

REPRODUCIBILITY STATEMENT

The repository includes cached GraphML, all generators and models, raw-output schemas, a 10,000-resample analysis script, and `reproduce_all.ps1`. Seeds are 11, 22, 33, 44, 55. The recorded environment is Python 3.13.11; NumPy 2.4.4; SciPy 1.17.1; NetworkX 3.6.1; pandas 2.2.3; scikit-learn 1.9.0; PyTorch 2.11.0 (CPU); Matplotlib 3.11.1; OSMnx 2.1.1; and GeoPandas 1.1.4. Hardware is an AMD64 Family 25 Model 68 system with 16 logical CPUs; CUDA was unavailable and no GPU was used. OSM data are copyright OpenStreetMap contributors under the ODbL [9].

APPENDIX A. END-TO-END EXPERIMENTAL PROTOCOL

For each seed, the pipeline executes the following fixed sequence:

- (1) generate connected synthetic base graphs and assign entire base-graph families to train, validation, or test;

- (2) generate equal numbers of independent and spatial-cluster failures;
- (3) compute intact λ_2 , u_2 , damaged λ_2 , the exact target, and the first-order estimate;
- (4) train direct-full, direct-no-Fiedler, direct-no-coordinate, residual-full, residual-no-Fiedler, residual-no-coordinate, Deep Sets, and summary-MLP models with identical partitions;
- (5) evaluate every model separately on both synthetic processes and on each OSM site-process pair; and
- (6) aggregate only after per-seed metrics have been calculated, then bootstrap seed indices and seed-paired differences 10,000 times.

This order is encoded in `run_paper_benchmark.py` and `analyze_paper_results.py`. The wrapper `reproduce_all.ps1` downloads or reuses OSM caches, runs seeds 11, 22, 33, 44, and 55, records the environment, and compiles this report.

APPENDIX B. OUTPUT AND CLAIM AUDIT

Machine-readable outputs include per-seed/per-mode metrics, bootstrap summaries, paired differences, error slices, runtimes, and environment versions. Raw model predictions and checkpoints are intentionally excluded from version control but can be regenerated. The public repository is <https://github.com/lee-vtruong/ResiliRoad>.

The evidence supports a narrow claim: an explicit residual spectral prior improves direct GCN calibration for the evaluated structural-disruption task. It does not support claims about flood probability, traffic service, city-wide robustness, or a universally necessary Fiedler node feature. This claim audit is included to keep poster-scale communication aligned with the actual experiment.

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